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Integrated OBN Acquisition Solutions

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Summary

This paper explores integrated Ocean Bottom Node (OBN) data acquisition solutions, focusing on overcoming the challenges posed by complex geological environments. It highlights how integrating advanced OBN technology with fit-for-purpose seismic vessels improves HSE performance, data quality and efficiency. The paper also discusses the design of OBN surveys, the specialized equipment used for the different terrains typically encountered in such surveys and the quality control measures implemented to ensure the integrity and quality of the high-resolution data acquired on seismic projects.

Key OBN Technologies

OBN Survey Design

The introduction of OBN has improved deployment flexibility and operation safety. This system can be deployed by rope, cable, or remotely operated vehicles (ROVs) and are designed to operate in water depths ranging from 0-4000m, even in densely populated infrastructure areas, which allows for greater adaptability to different survey conditions and improves the efficiency of data acquisition. OBN survey design is a sophisticated methodology to enhance the quantity and quality of seismic data collection. The push for innovation in OBN survey design has been driven by the challenges associated with subsurface complexity and necessity for accurate imaging solutions. These challenges have prompted the integration of advanced geological analysis.

The design process involves not only selecting appropriate shot or node interval but also considering logistical factors, such as the deployment of nodes and the timing of data acquisition. Each decision made during the planning phase can impact the effectiveness of the survey, making it crucial to balance operation efficiency with data quality. Below are four (4) tradition geometry types that are utilized in OBN project.

"Centre shooting" refers to a specific technique used to acquire seismic data with a single seismic source point located in the center of the survey area. In the center shooting configuration, the seismic vessel fires the seismic source in the center of the receiver spread. The nodes are spread out in a pattern around the source point, forming a polygonal layout. This acquisition method offers several advantages of improved data quality, better imaging of the center area, and enhanced data processing.

"Dual-sided shooting" refers to the technique used to acquire seismic data by deploying source on both sides of the survey area and the nodes are placed in between the source lines, offering the advantage of improved data coverage, reducing acquisition time.

"Single-sided shooting" is another seismic data acquisition technique used in offshore surveys. In this method, seismic sources are deployed on only one side of the survey area, while the nodes are placed on the other side. It is also commonly used in scenarios where one side of the survey area is inaccessible due to logistical or operational reasons.

"Cross-spread shooting" is a seismic data acquisition technique that involves the multiple intersecting source lines and receiver lines. Key features of cross-spread shooting include increased data redundancy. This redundancy aids in quality control, noise identification, and the reduction of acquisition-related artifacts.

In scenarios characterized by limited OBN resource availability, equivalent survey design methodologies offer a robust strategy for preserving target illumination and data quality. As shown in [figure 1](#), The core approach involves partitioning the nominal receiver carpet into two complementary receiver line sections. Critically, source lines are designed with significant overlap between the operational phases corresponding to each receiver line section. This overlapping shot ensuring that the aggregate fold, offset distribution, and azimuthal coverage across the combined dataset meet pre-survey objectives. For surveys with irregular operational boundaries, the zipper division line can be configured either as a straight line or, more strategically, aligned parallel to the operational boundary, as shown in [figure 2](#). This parallel alignment is generally preferred as it maximizes acquisition efficiency and minimizes the required node inventory.

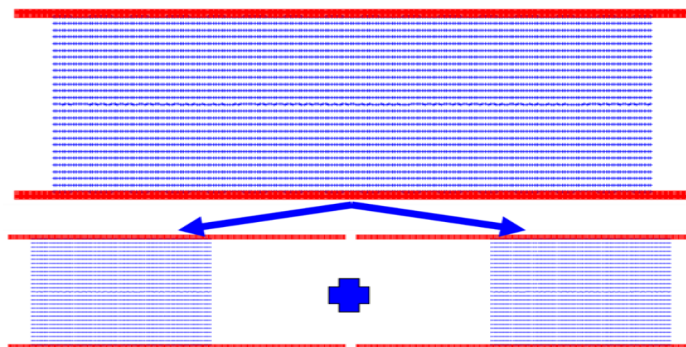


Figure 1—Equivalent survey design where receiver carpet is divided into two parts and shot lines are overlapped

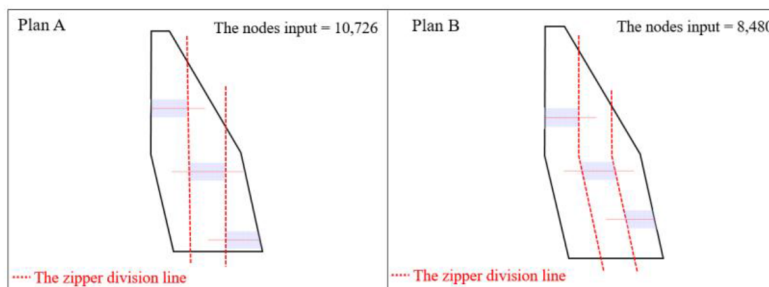


Figure 2—Zipper division methods with irregular operation boundaries

Seamless Acquisition Approach from Land to Sea

The seamless OBN acquisition is a significant advancement in marine seismic technology, enabling more efficient and higher-quality data collection in diverse environments. This approach integrates multiple processes and advanced technologies, ensuring optimal data quality while addressing the unique challenges posed by different underwater terrains and operational conditions. Consequently, this method provides

numerous advantages over conventional seismic acquisition methods, ultimately making it a preferred choice for many geophysical surveys.

Specifically, the seamless acquisition approach involves using different seismic sources and receivers across various surface conditions. As shown in [figure 3](#) is schematic diagram of seamless acquisition with different sources and receivers allocated.

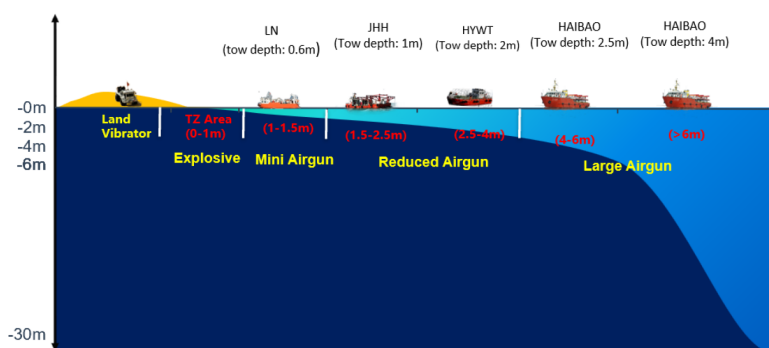


Figure 3—Seamless acquisition involving different sources and receivers

On the source side, explosive sources and vibrators are used on land. There are two types of vibrator, namely land vibrator and TZ vibrator. Land vibrators are deployed both onshore close to the sea and on the various islands in the survey area. In the more challenging terrain, a TZ vibrator was chosen as the optimum source solution. In offshore areas, deep-water air-guns, shallow water air-guns and mini air-guns are used according to the water depth. On the receiver side a variety of offshore nodes are deployed to record seismic data on the different terrains. Additionally, different types of node handling vessels and other specialized vessels, such as mini workboats and Hagglund (TZ transport vehicles) are used for the survey.

Particularly in areas with water depths of 0-1m—where neither vibrators nor air guns can be deployed—explosive sources are utilized. This method is also employed on surfaces covered by thick sabkha layers. In this area, shallow holes are drilled and a charge is lowered into the "drilling hole" and ignited. [Figure 4](#) shows the drilling equipment used on land and ultra-shallow water areas. In order to improve the drill efficiency, different drilling equipment is used depends on the complex surface conditions. Among these equipment, Haizuan and power rig can work in extreme shallow water up to 1m depth.



Figure 4—Different drilling equipment utilized in TZ project

Specialist Seismic Equipment

Integrated OBN acquisition solutions have revolutionized offshore seismic exploration by providing a comprehensive and highly effective approach to acquiring high-resolution data. By integrating advanced OBN with state-of-the-art seismic vessels, these solutions have significantly improved imaging capabilities, making them invaluable tools for the oil and gas industry's exploration and development efforts.

As shown in [figure 5](#) is BGP develop hybrid vessel and high deep water triple source vessel. BGP hybrid vessel is called BGP INOVATOR, which is the world's first large-scale dedicated DP seismic vessel possesses with integrated navigation, seismic shooting, node deployment and retrieval, and quality control capabilities, filling an industry gap with improved safety, environmental protection, and low carbon green standards for shallow water operations.



Figure 5—Hybrid vessel(left) and high-end deep water triple source vessel

The BGP high-end triple-source vessel features a DP-2 dynamic positioning system, accommodates 60 personnel, and operates 10 source strings to enable wide-tow triple-source configurations. Equipped with four compressors, the vessel ensures sufficient air supply volume up to 4050 cu.in. With a maximum shooting speed of 7.5 knots, this triple-source system significantly enhances data density compared to conventional dual-source acquisition methods. Consequently, survey efficiency increases by approximately 40%, while reduced towing duration mitigates equipment exposure to marine hazards.

The seamless OBN acquisition approach leverages modular nodal systems and automated handling techniques to streamline the deployment and retrieval processes. As shown in [figure 6](#) is the BGP modular node automatic deployment and retrieval system implemented in OBN project. The major features of this system is reliable and aotomatic node deployment and retrieval, which effectively reduces the manual operations and unsafe factors to ensure effective. Consist of automated robotic node handling systems, a battery-change system, a node-washing station, a data-docking unit, and a workshop, the modular system also provides other functions such as node storage, battery charging, and data downloading at the back deck.

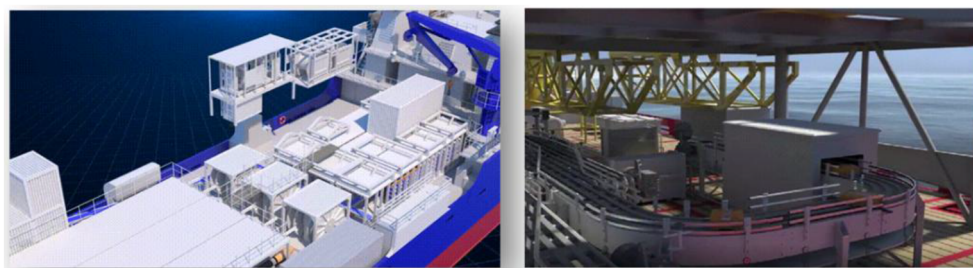


Figure 6—Modular node automatic deployment and retrieval system

The advances of umanned surface vehicles (USVs) revolutionize shallow-water exploration with centimeter-accurate GNSS-RTK navigation, enabling real-time node positioning in challenging 0-5m bathymetric zones through AI-powered obstacle avoidance and tidal current compensation systems. As shown in [Figure 7](#), flexible umanned surface vehicles (USVs) have been successfully deployed in OBN

projects. These USVs can be launched and recovered from node handling vessel. They communicate with the main vessel with radio system and use the same navigation system as the main vessel. Compatible with diverse vessel types and survey requirements, they support multiple deployment methods including ROV-assisted, tethered, or alternative techniques, and be able to execute ping operation according to pre-programmed survey plans, with the positioning accuracy comparable to conventional node-handling vessels.



Figure 7—Flexible unmanned surface vehicles (USVs)

Integrated Navigation System

Since 2009, DOLPHIN, a sophisticated navigation system developed independently by BGP for OBC / OBN operation, has provided multi-source vessel operation and navigation, node deployment and retrieval, positioning, multi-source vessel operation (Figure 8). Utilizing this navigation system and the VTS system we can monitor the production status of the vessel fleet in real time, and forewarn vessels operating close to production facilities and support vessels in transit in the survey area of any close passes or course adjustments required to ensure safe and secure operations in such high congested oil field production areas.

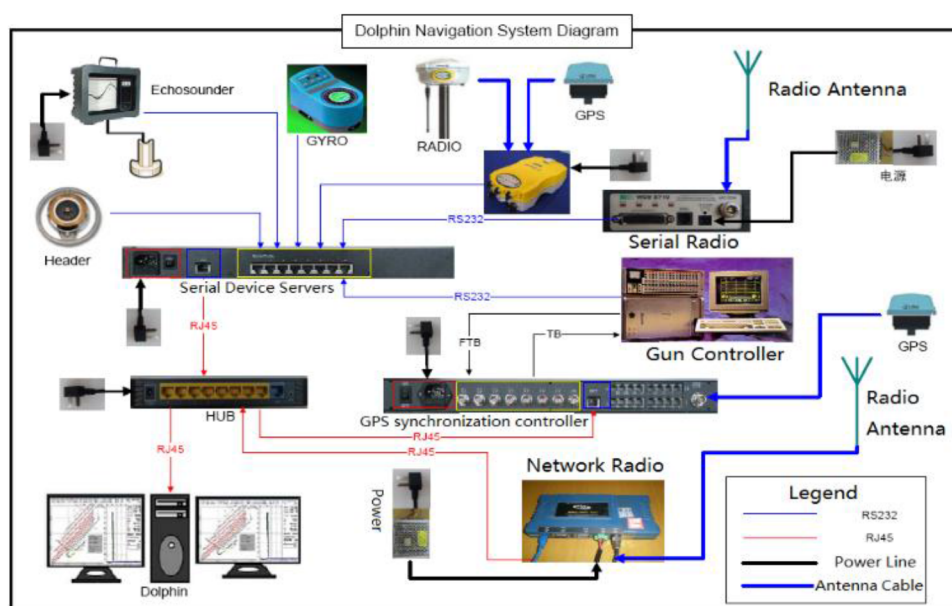


Figure 8—Diagram of Dolphin navigation and position system

Independent simultaneous shooting(ISS)

In OBN seismic exploration, increasing shot point density is a critical technical approach to improving seismic imaging quality. Independent simultaneous shooting(ISS) is a commonly used high-efficiency acquisition technique. In practice, operational methods such as single-vessel multi-source (Figure 9) and multi-vessel multi-source (Figure 10) configurations are widely adopted. Single-vessel multi-source acquisition involves a single vessel towing multiple seismic sources with an alternating firing mode, while multi-vessel multi-source acquisition entails coordinated operations with multiple air gun source vessels positioned at specific distance.

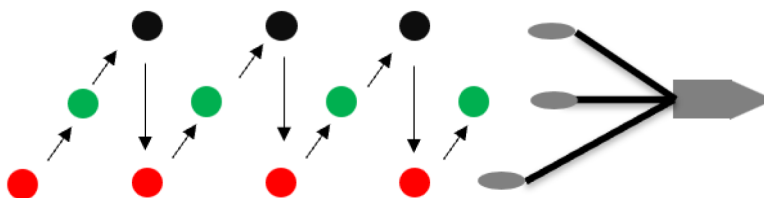


Figure 9—Schematic diagram of triple source shooting

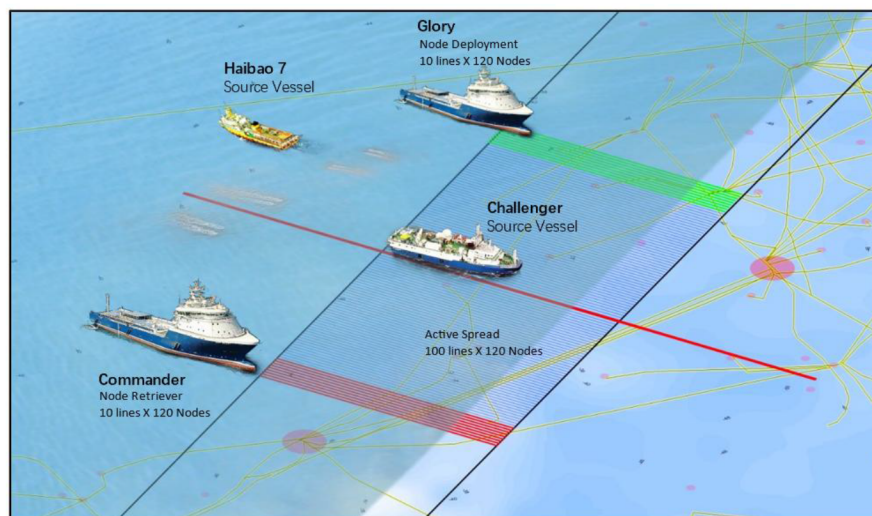


Figure 10—Diagram of multiple source Sim-Source shooting

Key parameters in ISS acquisition include random firing time, the distance between sources, and the speed of the source vessels. During air gun source firing, natural randomness in firing timing arises due to varying marine conditions such as tides and wind direction. By introducing controlled random dither times, the randomness of the firing time can be further enhanced, significantly improving data separation. Additionally, high efficient acquisition requires that there is no interference from adjacent sources in the clean record window. In single-vessel multi-source acquisition, this is achieved by controlling the vessel speed, while in multi-vessel multi-source acquisition, it is accomplished by managing the distance between source vessels. These measures ensure that the time interval between shot point exceeds the clean record length, thereby preventing interference in the target layer's reflection data. In the data processing phase, iterative inversion methods based on compressed sensing theory enable effective separation of blended data. This technical achieves a trade-off between seismic acquisition quality and operational efficiency.

Seismic Data Quality Control

With the improvement of acquisition efficiency in the last five years, the industry has witnessed a dramatic increase in the volume of OBN seismic data being generated daily with volumes in excess of ~15 Terabytes/

day in high-density acquisition operations. The dramatic increase in seismic data volume poses significant challenges to quality control, which encompasses a range of methodologies and practices aimed at ensuring the reliability and accuracy of seismic data collected through advanced OBN technology.

Figure 11 shows a comprehensive quality control (QC) work flow for OBN seismic data, focusing on time drift QC, node position QC, rotation QC and additional check such as DC analysis and RRY display (PRY represents the value recorded in the orientation sensor: pitch, Roll, and Yaw). The workflow integrates quantitative and qualitative methods to address challenges posed by variable bathymetry, complex subsurface geology.

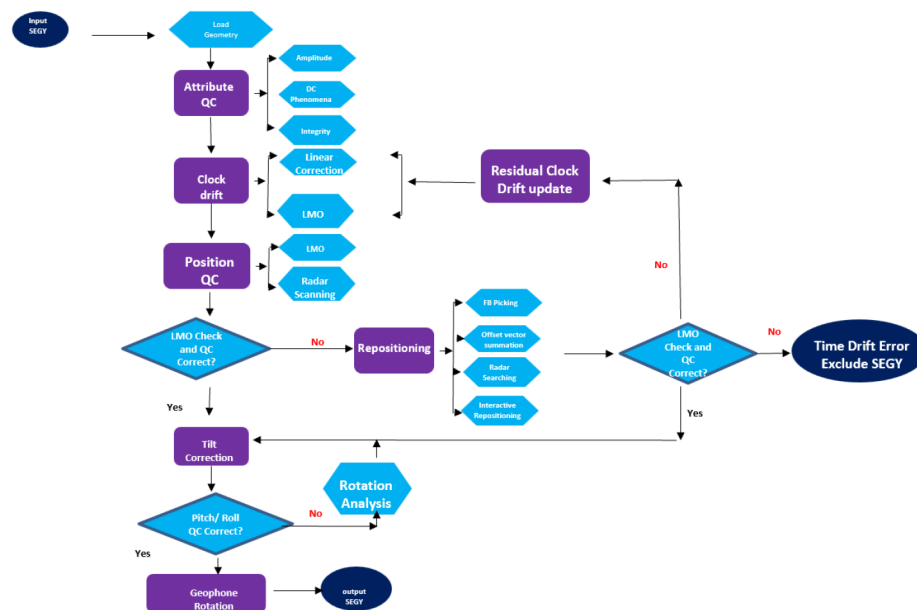


Figure 11—Comprehensive QC workflow for OBN seismic data

Specifically, attribute QC is used to observe abnormal amplitude, DC phenomena, or other node-related issue. Time drift QC leverages cross-correlation method to evaluate overall clock drift, comparing first arrival time against expected values from a reference station. Rotation QC enhances the reliability of multicomponent data, particularly for shear wave analysis. The basic principle is to confirm vertical geophone and IL/XL geophone have expected directional polarity. A failure in any of this warrant, additional analysis or manual scan is performed to refine orientation (Wang,2021).

Another import QC method is node position QC which is critical for accurate source-receiver geometry and reliable seismic imaging(Cao,2017). The QC workflow involves pre-deployment and post-deployment verification, automated analysis, and quantitative method to identify and assess deviations. Ensuring nodes are deployed according to the survey pre-plan is the first step in position QC. After deployment, node positions are validated against the pre-planned layout using acoustic positioning and first-arrival analysis to determine deviations caused by environmental and operational factors, such as variable bathymetry or ocean currents. The critical step of deviated node position QC involving getting the time when the node moved and the new position. The orientation sensor installed on the node records pitch, roll, and yaw (PRY) values every minute, enabling precise determination of the node's movement time. The new positions of nodes after movement can be determined using either acoustic ping operations conducted before retrieval or the data-driven First Break Picking (FBP) method.

Novel/additive information

From a design standpoint, surveys can be optimized within the limitation of budget, and operational constraints to meet the business objectives. Variable density survey is a new specialized methodology, involving the deliberate alteration of the density of shot points and/or the receiver points to enhance data acquisition and optimize geological interpretations. Notably, variable density designs improve the efficiency of seismic surveys, particular in challenging environments such as transition zone. Figure 12 illustrates an example of offset-depend variable shot density survey design. Specifically, it compares three configurations: (a) Design 1A, featuring a constant shot interval 25m;(b) Design 3A, where the shot interval is decreased to 50m beyond a crossline offset 1000m;(c) design 5A, which further reduce shot interval to 50m beyond a crossline offset of 675m. This survey geometry is optimized primarily considering the geological target is relatively shallow where sufficient near offset coverage is critical of near surface imaging. Optimizing shot density in far-offset reduces acquisition costs while maintaining sufficient data quality, achieving an effective balance between survey objectives and operational efficiency.

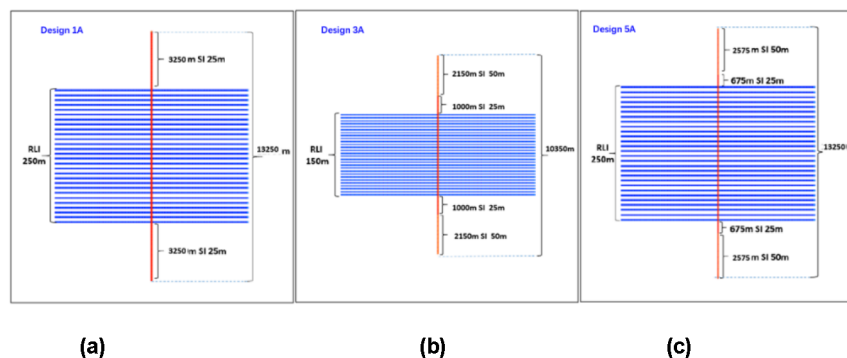


Figure 12—Variable shot density survey design. (a) Design 1A with constant shot interval 25m;(b) Design 3A with shot interval decreased to 50m beyond a crossline offset 1000m; (c) Design 5A with shot interval further decreased to 50m beyond a crossline offset 675m

Another important factor to consider is air gun array design in extreme shallow water. Traditional models may not adequately account for the modelled bubble radius, which is used to optimize the towed depth. To ensure the bubble formation and subsequent oscillation create a sound wave effectively, the bubble shall not reach the sea bottom or the sea surface. As demonstrated in figure 13, which illustrates the variation of air gun bubble radius with air gun volume, a clear inverse relationship is observed: reduced air gun volume yields smaller bubble radius. Consequently, arrays of smaller air guns are preferred for seismic operations in shallow water environments.

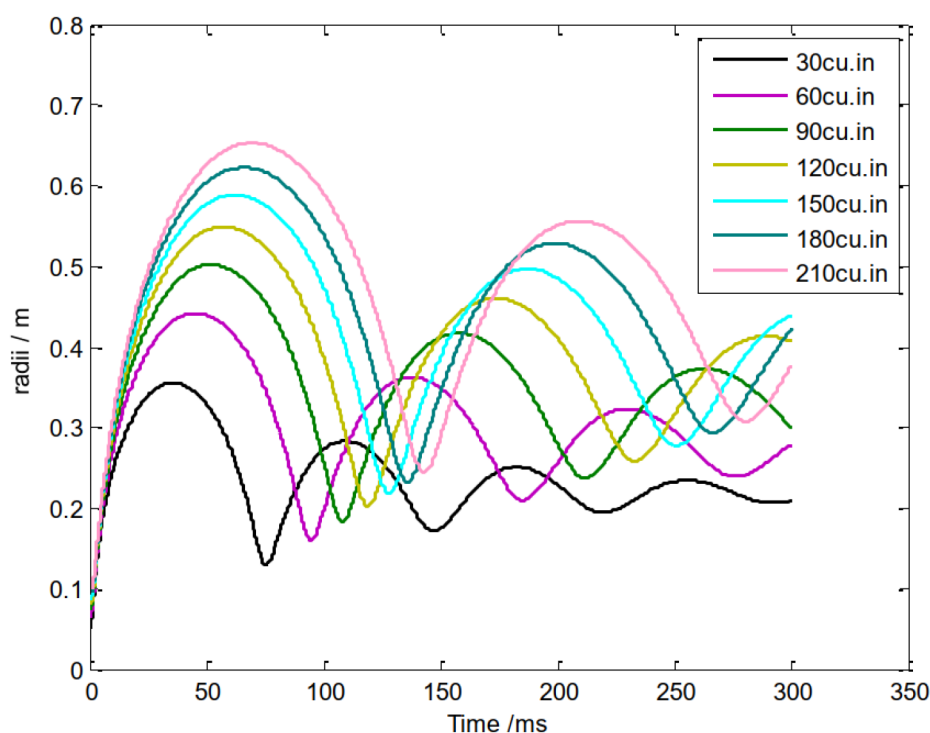


Figure 13—bubble radius variations with air gun volume at a fixed tow depth of 2 meters

By integrating Sim Source technology, production efficiency is significantly enhanced through the use of shallow-water dual-source vessels and multi-location node deployment. Figure 14 is an example of deployment of a node in two locations. The node leaves the GIM on the rack at T0 and starts recording immediately. The node is then deployed on position L1, P1 at time T1 and is retrieved at time T2, after which the node is re-deployed on position L2, P2 at time T3. Finally, it is retrieved at time T4. The node then goes back on the rack and connects to the GIM at time T5, and at that time recording stops. Using multi-position deployment allows the observers to deploy one node on several successive locations before going back on the rack for charging and harvesting. In comparison, with the traditional way the node needs to go back on the rack for harvesting and charging before it goes to another location. Using multi-position deployment can therefore save a lot of time in operations. The nodes need not go back to the GIM so often. This works for any number of locations as long as there is enough power in the battery.

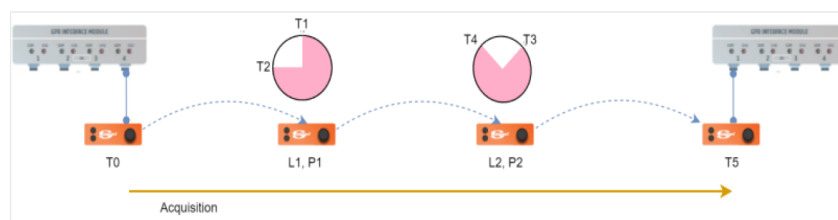


Figure 14—Multi-position node deployment

Case Studies

To illustrate the application of the integrated OBN solution, three case studies are given as below.

The first case is BP Indonesia project acquired in 2017-2018, which is the world's first large-scale high-density OBN acquisition project covering over 800 km² which was successfully acquired. Challenges included multiple oilfield platforms and currents of up to 3 knots. A full-azimuth, high-fold, large-offset geometry was designed with a shot density of up to 5 million per km² using ISS acquisition, significantly

improving acquisition efficiency while producing excellent data quality. Figure 15 is the geometry parameter and figure 16 is the imaging comparison between legacy data and current acquisition data.

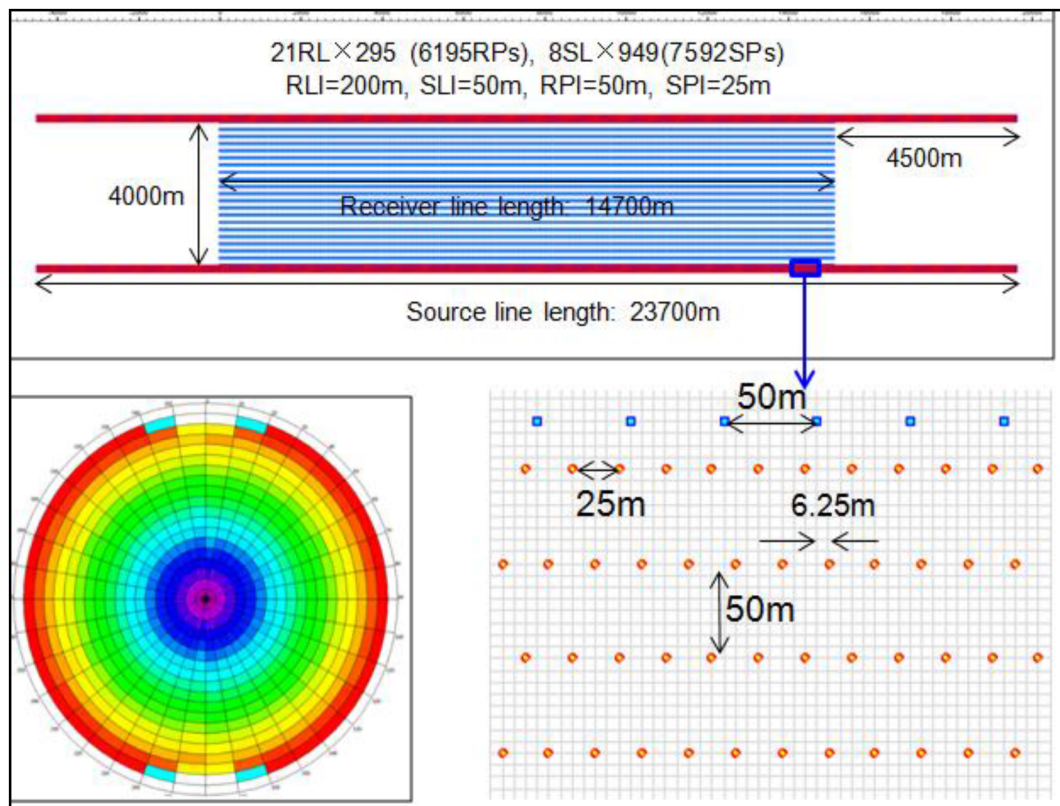


Figure 15—Geometry of BP, Indonesia project

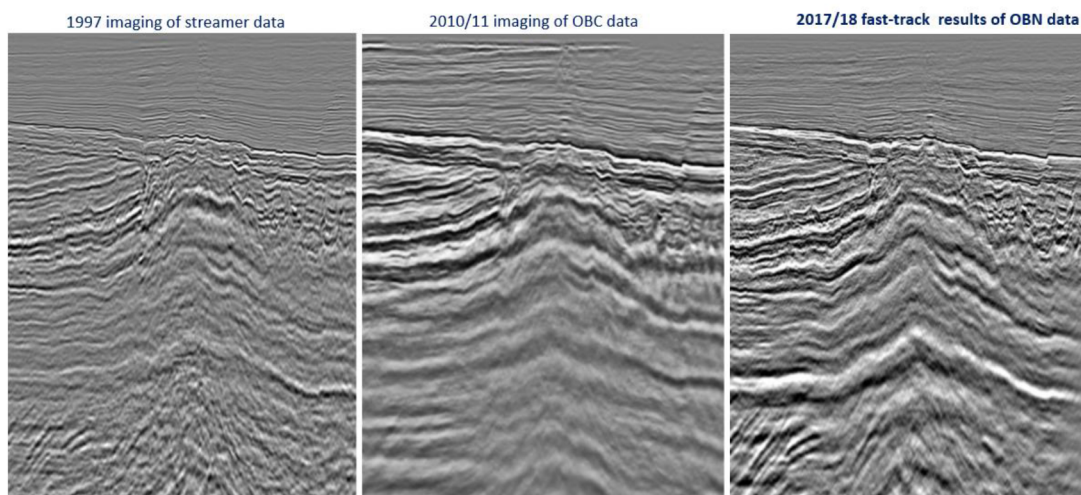


Figure 16—Imaging comparison between legacy data and current acquisition data: imaging of streamer data acquired in 1997 (left); imaging of OBC data acquired in 2010 (middle); and fast-track results of OBN data from Indonesia project acquired in 2017 (right).

The second case is Brunei Shell project acquired in 2019-2000 (Figure 17), which is a combined NOAR/ROV OBN survey with high shots and receiver density. The challenge includes highly sensitive seafloor environment with protect coral reefs, multiple pipelines and shipwrecks, and monsoon season interference

with 35-40 knot winds. Integrated OBN acquisition techniques and equipment are applied to this project to ensure the operation efficiency and data quality.



Figure 17—Brunei Shell OBN project overview

The third case is UAE ADNOC project which is the world's largest integrated seismic project. As shown in figure 18, the project covers an area of 63,000 km² 3D mega-survey consisting of 3 distinct areal subsets, each requiring their own specialist equipment and acquisition techniques, in which offshore OBN area covers 30,000 km² with a 2500 km² subset over Zakum oilfield, and TZ OBN area covering 10,000 km². 4 crews were working simultaneously and more than 50 seismic vessels were involved in the operations. 32,000 ocean bottom nodes were used in the project and over 1,000 personnel were involved in the operations.

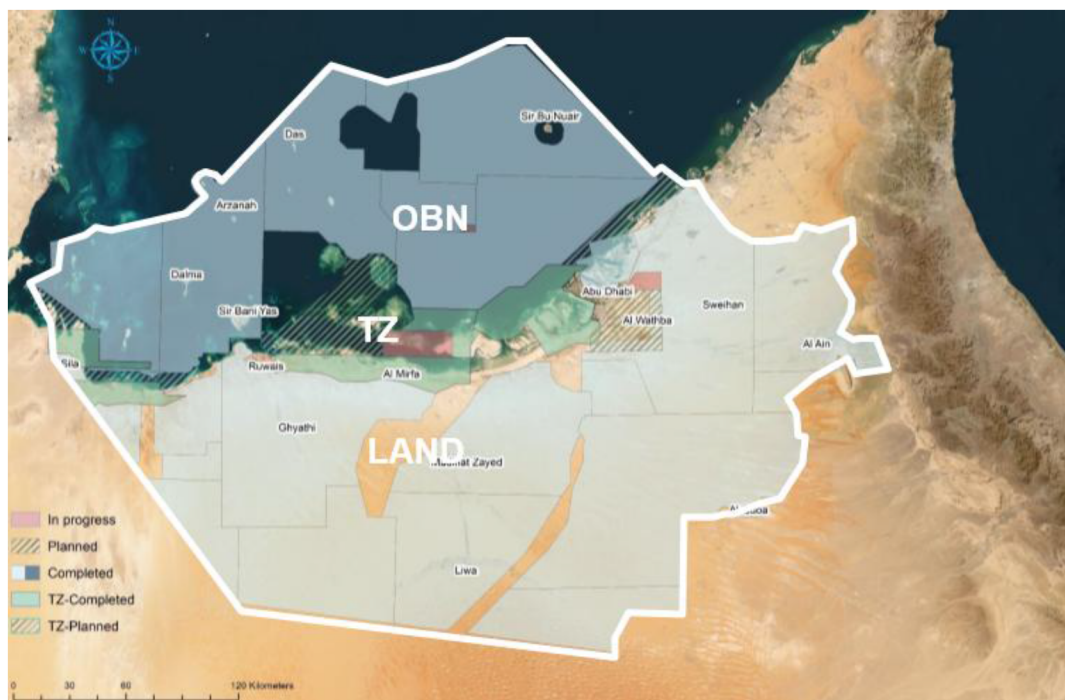


Figure 18—UAE ADNOC project overview

Conclusions

Integrated Ocean Bottom Node (OBN) acquisition solutions have demonstrated significant improvement in data quality, operational efficiency, and safety on large-scale surveys. By incorporating advanced survey designs, multi-source shooting and specialized equipment, these solutions have effectively overcome the key challenges of complex geological conditions, sensitive environments and adverse weather conditions. OBN technology facilitates the acquisition of high-resolution data more efficiently which substantially reduces survey durations. This, in turn, has resulted in accelerated delivery of high-fidelity seismic imaging – so critical for better interpretation of the subsurface and optimize decision-making in offshore exploration. BGP's OBN technology proven to be a significant game changer in overcoming the challenges inherent for shallow water surveys especially in the transition zone, with ongoing innovation ensuring even greater potential for future seismic applications. The continued development of these solutions promises to deliver even greater value to customers in the energy industry.

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